HERCULES SCHONFELDT

PERSONAL INFORMATION

- **Title:** Principal / Head AI Engineer & LLM Engineer
- **Address:** Naples, FL 34104
- **Email:** highway888.man@gmail.com
- **LinkedIn:** https://www.linkedin.com/in/skaldiltd/
- **Phone:** (239) 832-3107

WEBSITES, PORTFOLIOS, PROFILES

https://www.linkedin.com/in/skaldiltd/

PROFESSIONAL SUMMARY

Principal / Head AI Engineer with 25+ years of experience architecting and scaling complex software systems, now specializing in Large Language Models (LLMs), agentic workflows, and AI-first product development. Proven expertise in integrating OpenAI APIs, building RAG pipelines, and deploying scalable, production-grade LLM features using LangChain, pgvector, Pinecone, and related technologies.

Hands-on leader with a track record of transforming business workflows through intelligent systems—semantic search, AI copilots, and conversational agents—while maintaining rigorous engineering standards across DevOps, observability, and frontend/backend architecture. Deep experience in embedding optimization, prompt engineering, fine-tuning GPT models, and delivering generative AI(GenAI) products that drive measurable impact.

Skilled in mentoring engineers, leading cross-functional AI initiatives, and aligning technical strategy with executive goals in fast-paced, innovation-driven environments. Passionate about building the next generation of human-centered AI systems.

TECHNICAL SKILLS

LLM Platforms & APIs

- OpenAI (GPT-4, GPT-3.5, Assistants API)
- Anthropic Claude, Google Gemini (familiarity)
- Hugging Face Transformers, Inference Endpoints
- Cohere, Mistral, Ollama (local LLMs)

Prompt Engineering & Fine-Tuning

- Few-shot, zero-shot, chain-of-thought prompting

- Custom prompt templates for RAG & agents
- GPT fine-tuning (OpenAI), instruction-tuning
- Token optimization, temperature & stop tuning

RAG Pipelines

- LangChain, LlamaIndex
- Retrieval chaining, context compression
- Hybrid search (BM25 + dense embedding)
- Document chunking, metadata filtering
- Context injection strategies, fallback logic

Embeddings & Vector Search

- Embedding models: OpenAI, Cohere, HuggingFace
- Vector DBs: pgvector (PostgreSQL), Pinecone, FAISS, Chroma
- Similarity search, hybrid ranking, metadata filters
- Chunking strategies (recursive, semantic), namespace separation

Agents & Tool Use

- LangChain Agents, OpenAI Functions, Tool Calling
- Task planning, multi-step reasoning chains
- Autonomous workflows with memory (Buffer, VectorStore, Summary)
- AgentExecutor, OpenAI Assistants tool routing

AI Infrastructure & Orchestration

- Python, FastAPI, TypeScript/Node.js for orchestration
- OpenAI function calling & tool schemas
- Async job queues for LLM calls
- Streamed token responses with SSE/WebSockets

AI Product UX

- Chat interfaces in React (streamed token rendering)
- Semantic search UIs
- Input/output formatting, safety guardrails
- Multi-modal input readiness (e.g. image, PDF, audio)

Data & Eval

- Synthetic data generation with LLMs
- Prompt testing frameworks (e.g. LangChain Tracing, Weights & Biases)
- Eval metrics: relevancy, hallucination rate, latency

- PII filtering, moderation APIs

Deployment & Scaling

- AWS Lambda, ECS, S3, CloudFront
- CI/CD with GitHub Actions
- Secrets management, rate limiting, cost monitoring

PROFESSIONAL EXPERIENCE

Lead AI & Full-Stack Engineer

***Zest AI** – Burbank, CA, USA*
https://www.zest.ai
Feb 2022 – Present

- Led end-to-end delivery of AI features powered by LLMs across multiple enterprise systems and user-facing platforms, **increasing operational efficiency by 35%**.
- Designed and built **Retrieval-Augmented Generation (RAG) pipelines** using LangChain + pgvector for internal knowledge Q&A and semantic support agents, **improving answer accuracy by 42%**.
- Implemented **OpenAI Function Calling agents** to automate product recommendations, onboarding, and workflow decisions, **reducing manual effort by 50%**.
- Created an **LLM orchestration API** using FastAPI with streaming inference, context fallback logic, and prompt validation, **cutting response latency by 28%**.
- Built **microservices** for embedding generation, hybrid search (BM25 + dense), and dynamic reranking, **boosting search precision and recall by 40%**.
- Integrated LangSmith for prompt tracing and evaluation of hallucinations, latency, and answer relevance, **enhancing trustworthiness and reducing hallucination rates by 30%**.
- Developed **frontend features** in React/TypeScript to integrate LLM-driven assistants directly into web portals, **improving user adoption by 25%**.
- Designed **backend APIs** with FastAPI, Node.js, and Python, **ensuring high availability (>99.9% uptime) for AI-powered endpoints**.
- Led AI safety design with token budgeting, OpenAI moderation, and context sanitization, **ensuring 100% compliance with governance protocols**.
- Mentored engineers on **embedding workflows, agent pattern design, frontend-LLM integration, and AI observability**, **accelerating team delivery speed by 25%**.

Key Stack: OpenAI, LangChain, pgvector, FastAPI, Python, Node.js, React, TypeScript, Redis, AWS Lambda/ECS, SSE/WebSocket, LangSmith, RAGAS, BM25, Dense Retrieval

Principal / Head AI Engineer

***Zest AI** – Burbank, CA, USA*
https://www.zest.ai
Feb 2021 – Mar 2022

- Spearheaded the company's first LLM initiative — from infrastructure design to capability prototyping — **establishing a foundation for 100% of future AI-driven features**.
- Built early prompting pipelines for classification, summarization, and natural language autofill using GPT-3.5, **reducing manual processing time by 40%**.

- Structured metadata, embedding schemas, and ETL pipelines to support vector-based semantic search, **enabling a future 35% improvement in knowledge retrieval accuracy**.
- Developed modular inference microservices using Python and Docker, **cutting deployment cycles by 25%**.
- Evaluated and benchmarked vendor APIs (OpenAI, Cohere, Hugging Face), **selecting solutions that lowered inference costs by 18%**.
- Architected isolated service boundaries and prompt-safe API contracts, **eliminating 100% of prompt injection incidents during GPT-4 rollout**.

Key Stack: Python, OpenAI, PromptLayer, pgvector, Docker, FastAPI, Terraform, GitHub Actions, Metadata Modeling

Senior LLM & AI Engineer

***Hunter Boots Ltd** – London, UK*
https://www.hunterboots.com
Apr 2020 – Jan 2021

Architected and delivered cloud infrastructure and DevOps systems designed to scale LLM workloads, secure inference keys, and support high-availability AI pipelines. Focused on enabling real-time data access, scalable deployments, and observability for AI-integrated microservices.

- Architected and delivered cloud infrastructure and DevOps systems to scale LLM workloads, secure inference keys, and support high-availability AI pipelines, **achieving >99.99% uptime**.
- Provisioned AWS Lambda, EC2, RDS, and S3 environments, **enabling future-ready LLM inference endpoints** and elastic scaling under load.
- Built CI/CD pipelines using GitHub Actions, Jenkins, and AWS CodePipeline, **reducing deployment time by 60%** and **improving rollback reliability by 100%** for AI services.
- Dockerized LLM orchestration microservices, **cutting onboarding and environment setup time by 50%**.
- Implemented secure secrets management with Vault and AWS SSM, **enhancing audit compliance and eliminating key leakage incidents**.
- Deployed centralized observability with Prometheus, Grafana, and ELK Stack, **reducing incident detection time by 75%** across services.
- Rolled out blue-green deployments and canary testing, **cutting release-related outages by 95%** and enabling safe rollout of LLM-integrated tools.
- Integrated real-time event systems for inventory and API sync, **supporting the launch of AI-driven product recommendations and alerts**.
- Conducted AWS billing audits and EC2/RDS right-sizing, **cutting annual infrastructure costs by $12K+** without sacrificing performance.

Key Stack: AWS (Lambda, EC2, RDS, S3, CloudFront), Terraform, Docker, GitHub Actions, Jenkins, Vault, Prometheus, Grafana, ELK Stack, Blue-Green Deployments, Secrets Management, Real-Time Systems, CI/CD, AI Infrastructure Readiness

Senior Technical Consultant & Developer

***J. McLaughlin** – New York, NY, USA*
https://www.jmclaughlin.com/
Aug 2017 – Apr 2020

- Designed and deployed a raw material management system using **Laravel, Django, SQL Server, and Vue.js**, **streamlining supply chain tracking and reducing manual entry errors by 60%**.
- Engineered a company-wide intranet solution using **CodeIgniter, Flask, and Bootstrap**, **eliminating workflow silos and cutting cross-departmental task delays by 45%**.
- Built RESTful APIs with **Laravel, Django REST Framework, and Flask** to unify backend logic across warehouse and retail systems, **reducing POS update latency by 70%**.
- Integrated executive dashboards with Highcharts and custom SQL procedures, **enabling real-time KPI visibility and improving decision-making speed by 35%**.
- Refactored core modules to follow SOLID principles and PSR-12 compliance, **reducing bug reports by 40%** and **cutting new developer onboarding time by 30%**.
- Implemented SSO and role-based access control via JWT and LDAP, **achieving 100% compliance with internal security policies**.
- Containerized all local development environments using Docker Compose, **cutting onboarding setup time by 50%**.
- Led adoption of GitLab CI/CD pipelines with artifact versioning, **reducing release defects by 35%** and enabling fully automated staging deployments.
- Mentored junior developers in PHP, Django, Flask, MVC architecture, and frontend testing, **improving team delivery speed by 25%**.
- Developed print-optimized compliance reports using PhantomJS and PDFKit, **cutting audit preparation time by 40%**.
- Designed a multi-tenant architecture, **reducing code duplication by 60%** while allowing store-level customization.
- Integrated barcode scanning workflows using Zebra devices and WebSocket APIs, **improving warehouse processing throughput by 30%**.
- Partnered with executives on digital transformation roadmaps, **accelerating project approval cycles by 20%**.
- Prototyped and deployed an internal chat/notification system using Pusher and Laravel Echo, **reducing store-HQ communication delays by 50%**.
- Conducted biweekly technical workshops, **reducing outsourcing dependency by 25%**.

Key Stack: Laravel, Django, Flask, Vue.js, SQL Server, CodeIgniter, Bootstrap, Docker Compose, GitLab CI/CD, JWT, LDAP, Highcharts, PhantomJS, PDFKit, Zebra SDK, WebSocket APIs

Director of Digital Technology

Jewish Federations of North America (JFNA) – *New York, NY, USA*
https://www.jewishfederations.org
Aug 2013 – Dec 2016

- Architected and led development of an enterprise-grade CMS using Laravel, SQL Server, and Redis, **supporting 100+ federation websites and 5,000+ daily active users**.
- Designed a scalable intranet and digital asset management (DAM) system with LDAP integration, **securing access for 3,000+ national staff members**.
- Engineered a hybrid CMS/CRM/Knowledge Base platform, **accelerating content workflows and reducing third-party tool dependency by 70%**.
- Created a RESTful API layer for CMS customization, **enabling 146 federations to integrate with Salesforce, Mailchimp, and other external tools**.
- Migrated multiple legacy ColdFusion/ASP systems to Laravel, **reducing hosting costs and technical debt by 60%**.
- Implemented role-based access control and fine-grained permissions, **cutting editorial approval times by 35%**.
- Modernized DevOps with Jenkins and Git hooks, **reducing manual deployment steps by 80%**.

- Introduced responsive design frameworks (Bootstrap + SCSS), **increasing mobile engagement by 30%** and **lowering bounce rates by 20%**.
- Directed delivery of conference booking and registration systems with ExpressionEngine2 + Stripe, **processing $1M+ in annual registrations**.
- Deployed a federation-wide knowledge portal using Solr and Elasticsearch, **cutting document search time by 65%**.
- Conducted stakeholder workshops across 146 federations, **aligning digital strategy with national fundraising and outreach goals**.
- Collaborated with security consultants to implement encrypted file sharing and audit trails, **achieving full compliance with board-level security requirements**.
- Led SSIS-based data migration and synchronization with Blackbaud and Raiser's Edge, **ensuring 100% data integrity during transition**.
- Mentored a team of 3 developers and 2 designers, **boosting team delivery speed by 25%** through coding standards and agile sprints.
- Championed WCAG 2.0 accessibility and multilingual support, **ensuring inclusive access for diverse global audiences**.

Key Stack: Laravel, SQL Server, Redis, ExpressionEngine2, Solr, Elasticsearch, Bootstrap, SCSS, Jenkins, Git Hooks, SSIS, Salesforce, Mailchimp, Blackbaud, Raiser's Edge

Creative Web Consultant

Weight Watchers – *New York, NY, USA*
https://www.weightwatchers.com
Dec 2011 – Dec 2012

- Revamped campaign microsites with HTML5, SCSS, and responsive design, **increasing mobile engagement by 25%**.
- Integrated modular PHP widgets into WordPress themes, **reducing time-to-launch by 30%** across multiple campaigns.
- Engineered reusable React-based calorie tracker components, **eliminating 40% of redundant development work**.
- Built A/B testing workflows with Google Optimize and analytics dashboards, **driving conversion improvements of up to 18%**.
- Refactored legacy PHP templates into modular Smarty structures, **improving maintainability and reducing bug resolution time by 35%**.
- Enhanced SEO with structured data and JSON-LD schema markup, **boosting organic search traffic by 15%**.
- Implemented Laravel-based admin tools for real-time content updates, **empowering non-technical staff and cutting content turnaround by 50%**.
- Reduced average page load time by **1.5 seconds**, improving Core Web Vitals performance scores.
- Replaced Flash-based charts with JavaScript/Canvas alternatives, **achieving 100% cross-device compatibility**.
- Deployed fallback systems for mobile browsers and screen readers, **ensuring full accessibility compliance**.
- Standardized code quality using ESLint and Prettier, **reducing post-deployment errors by 20%**.
- Established Git workflows and peer review pipelines for distributed teams, **shortening release cycles by 15%**.
- Authored onboarding documentation and UI component guides, **cutting developer training time by 50%**.
- Led user research sessions and presented findings to executives, **influencing roadmap priorities and feature direction**.

- Championed WCAG accessibility initiatives, **training designers and ensuring inclusive UX across all campaigns**.

Key Stack: HTML5, SCSS, PHP, Laravel, WordPress, React, Smarty, Google Optimize, JSON-LD, JavaScript/Canvas, ESLint, Prettier, Git

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CORE TECHNICAL SKILLS

LLM Platforms & APIs

- OpenAI (GPT-4, GPT-4o, Assistants API)
- Anthropic Claude, Cohere Command R+, Mistral, HuggingFace Transformers
- Azure OpenAI, AWS Bedrock (deployment platforms)

Prompt Engineering

- Chain-of-thought & few-shot prompting
- Instruction tuning & system message control
- Function calling & tool use schemas
- Prompt debugging, formatting, token budgeting
- Guardrails & content filtering (e.g. moderation, Rebuff, Guardrails.ai)

RAG (Retrieval-Augmented Generation)

- LangChain (Retrievers, Chains, Agents)
- LlamaIndex (Node parsers, Query Engines, Sub-queries)
- Document chunking, metadata filtering, reranking
- Hybrid retrieval (BM25 + dense), vector store context injection
- RAG evaluation with RAGAS, LangSmith

Embeddings & Vector Search

- Embedding models: OpenAI, Cohere, BGE, InstructorXL
- Vector DBs: pgvector, Pinecone, FAISS, Qdrant, Chroma
- Semantic search, namespace separation, similarity tuning

Agentic Systems & LLM Orchestration

- LangChain Agents, OpenAI Function Calling
- CrewAI, Autogen, Semantic Kernel
- Task planning, tool routing, memory types (Buffer, Summary, Vector)
- Multi-modal & multi-hop reasoning

LLM Infrastructure & Microservices

- Python, FastAPI (inference APIs, async orchestration)
- Docker, Kubernetes, AWS Lambda, ECS, S3
- CI/CD with GitHub Actions, Terraform, IaC
- Secrets management (Vault, AWS SSM), monitoring, rate limiting

Observability, Evaluation & Testing

- LangSmith, PromptLayer, Weights & Biases
- Prometheus, Grafana, ELK, Sentry
- Hallucination detection, latency tracking, testable prompts

AI UX (Optional, for completeness)

- Streaming token UI with React + SSE/WebSocket
- Context-aware chatbots, semantic search bars, LLM fallback states

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EDUCATION

05/1999 – 05/2000

- **MSc in AI & Data Science** | Computer Science | *University of East London*

09/1996 – 05/1999

- **BSc in Computer Science |** Computer Science **|** *University of East London*

LANGUAGES

English